

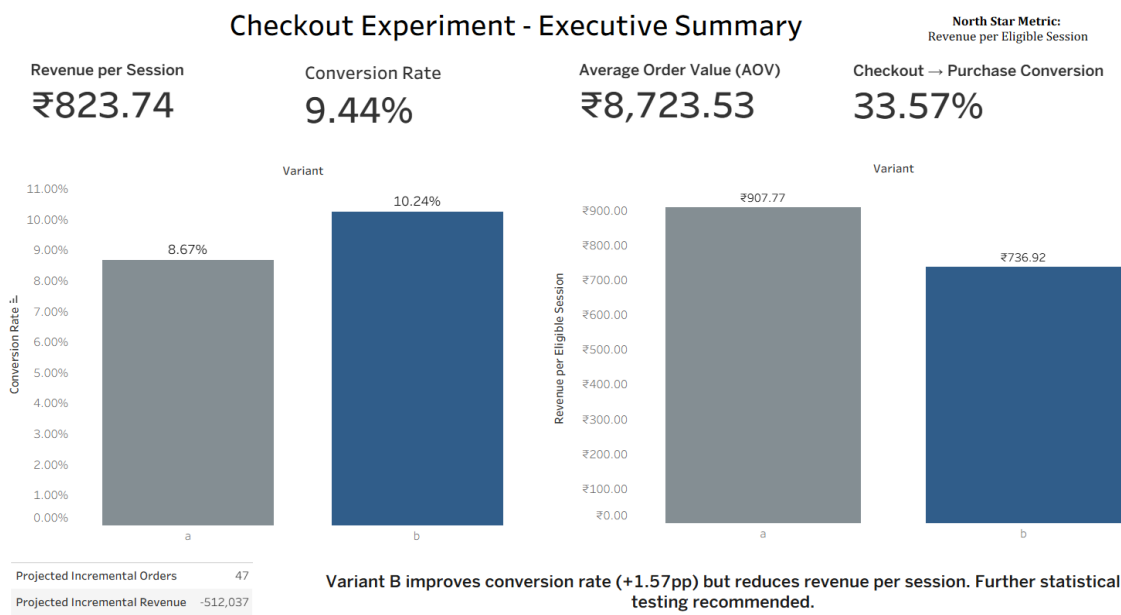
Checkout Experiment – Final Decision Memo

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1. Business Goal & KPIs

Objective: Evaluate whether Variant B should be rolled out platform-wide by assessing conversion uplift, revenue impact, margin proxy, and segment-level performance.

- North Star: Revenue per Eligible Session
- Conversion Rate (Eligible → Purchase)
- Funnel Step Conversions
- Checkout-to-Purchase Conversion
- Revenue per Session
- Average Order Value (AOV)
- Discount Rate & Margin Proxy
- Payment Progression Rate



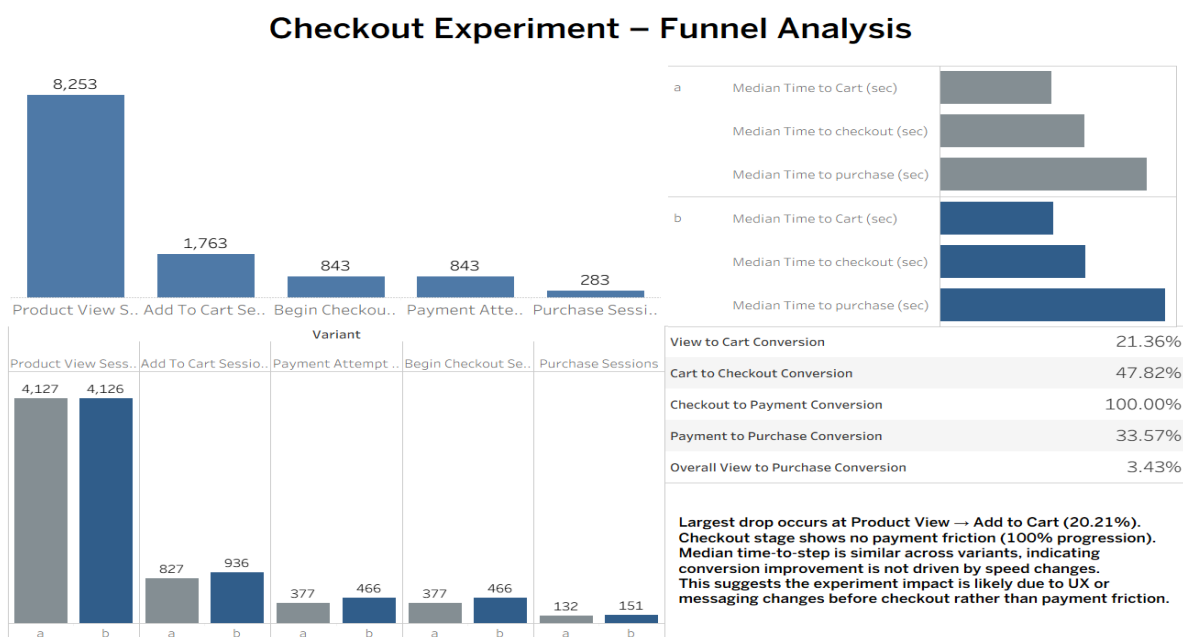
2. Data & Methodology Summary

Built reproducible ETL pipeline generating fact_sessions, fact_orders, dim_users_enriched. Applied deduplication, casing normalization, missing device/channel handling, IQR-based outlier capping, margin proxy enrichment, and time-to-step metrics. Eligibility defined as product-view sessions within experiment window.

3. Key Insights

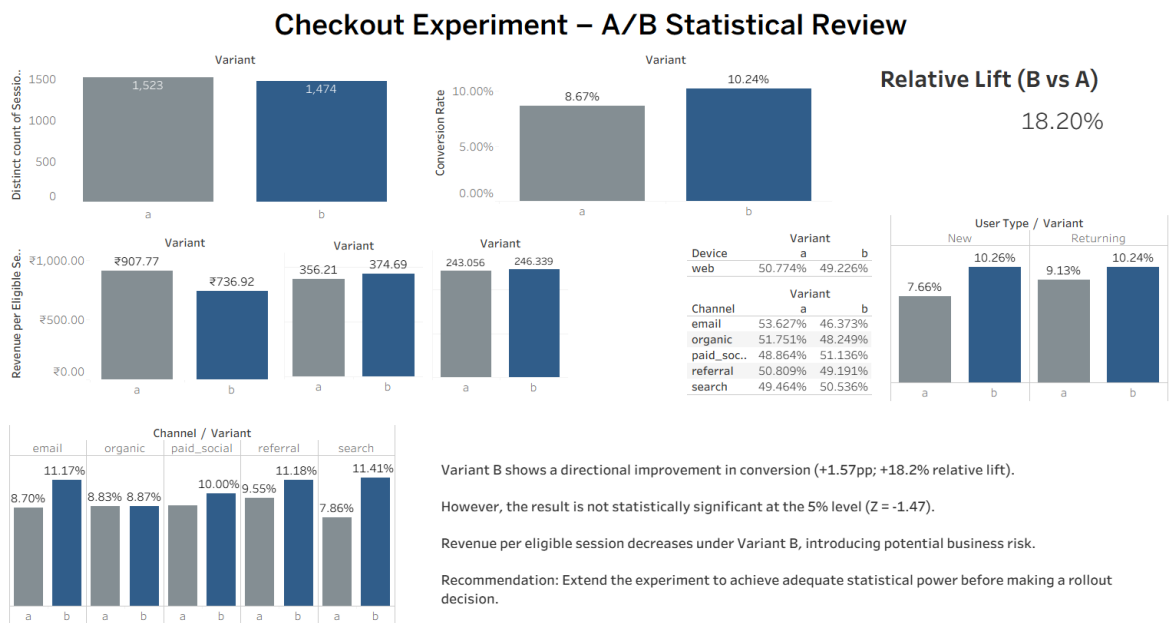
- Conversion improved from 8.67% to 10.24% (+1.57pp; +18.2% relative lift).
- Revenue per session declined under Variant B.
- Largest funnel drop: Product View → Add to Cart (20.21%).

- Checkout → Payment shows no friction (100% progression).
- Premium & Search segments show strongest uplift.
- Tier 2 cities show highest relative improvement.
- Revenue decline concentrated among Value users.



4. Experiment Outcome & Statistical Validation

Two-proportion Z-test: Directional lift but not statistically significant at 5% level ($Z=-1.47$). Confidence interval overlaps zero. Sample size insufficient for decisive inference.



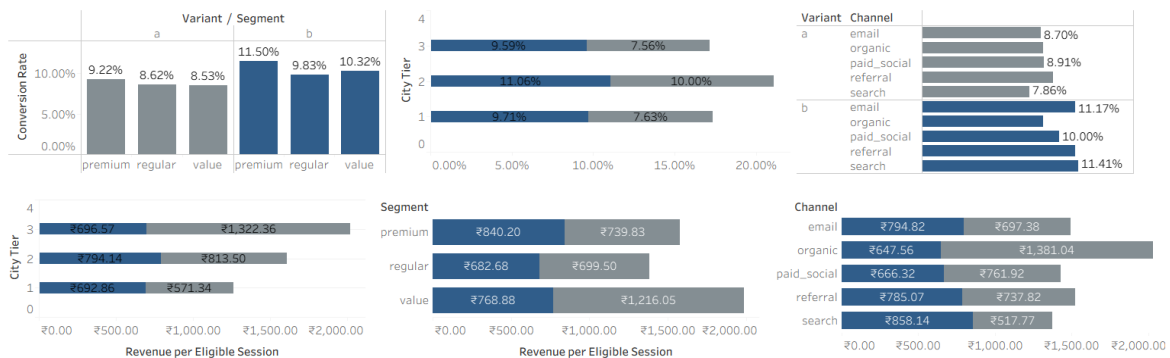
5. Segment Findings (HTE-lite)

Variant B benefits Premium users and Search traffic most. Neutral/slight negative impact for Value users. Tier 2 cities demonstrate strongest relative lift.

Checkout Experiment – Segment Explorer

Performance by User Segment, Channel & City Tier

Variant
☒ a
☒ b



Variant B delivers strongest conversion lift among Premium users and Search traffic. However, revenue per eligible session declines across most segments, particularly for Value users. Tier 2 cities exhibit the highest relative conversion improvement.

6. 30-Day Impact Estimate

- Projected incremental orders (base case): +47
- Revenue impact potentially negative due to lower RPS
- Sensitivity modeled: Conservative / Base / Optimistic scenarios
- Margin proxy impact indicates short-term profitability risk

7. Recommendations (MoSCoW Prioritization)

- Must: Extend experiment to achieve adequate power.
- Must: Investigate revenue erosion among Value users.
- Should: Optimize View → Cart UX friction.
- Should: Run targeted test for Premium & Search users.
- Could: Refine pricing/discount mix to protect margin.

8. Risks, Limitations & Follow-up Experiments

- Statistical insignificance limits rollout confidence.
- Revenue decline may offset conversion gains.
- Segment heterogeneity introduces uneven impact.
- Experiment duration may not capture long-term behavior.

Final Decision: Do NOT roll out Variant B at current stage. Extend testing, improve monetization, and re-evaluate with sufficient power.